Marmara University – Electrical and Electronics Engineering - STAT2056 Midterm - 3/5/2023, 17:00 (Duration 75 minutes)

Question 1 (20 pts)

Police plan to enforce speed limits by using radar system at four different locations L1, L2, L3, L4 within the city limits. The radar system at each of these locations are operated for 40%, 30%, 20% and 10% of the time. If a person who is speeding on his way to work has probabilities of 0.2, 0.1, 0.5, and 0.2 respectively of passing through these locations,

(a) (10 pts) What is the probability that he will be fined?
(b) (10 pts) If the person received a speeding ticket on her way to work, which radar location was most probable that she passed through?

Question 2 (35 pts)

(a) (10 pts) Determine the value of c so that the following function can serve as a probability distribution of the discrete random variable X.

$$f(x) = c\binom{2}{x}\binom{3}{3-x} \quad x = 0,1,2$$

(b) (10 pts) Compute the mean and variance of X.
(c) (10 pts) Plot the probability mass function and cumulative
(d) (5 pts) Compute the mean and variance of Y = 2X+1.

Question 3 (25 pts)

The probability that a student at a local high school fails the screening test for scoliosis (curvature of the spine) is known to be 0.001. Of the next 1900 students at the school who are screened for scoliosis, approximate the probability using Poisson distribution ($f(x) = \frac{e^{-\alpha}\alpha^x}{x!}$ where $E[X] = \alpha$) that

(a) (8 pts) None of the students fail the test.
(b) (8 pts) at most 2 fail the test;
(c) (7 pts) Exactly 4 students fail the test.

Question 4 (20 pts)

A scientist inoculates mice, one at a time with a disease germ until he finds 2 that have contracted the disease. If the probability of contracting the disease is 1/6,

(a) (10pts) What is the probability that 8 mice are required?
(b) (10pts) What is the probability that 8th mouse is the 1st one with a disease germ?

STAT 2056 Midterm Solutions 3/5/2023

Q1 $P(L_1) = 0.2$ $P(R|L_1) = 0.4$
$P(L_2) = 0.1$ $P(R|L_2) = 0.3$
$P(L_3) = 0.5$ $P(R|L_3) = 0.2$
$P(L_4) = 0.2$ $P(R|L_4) = 0.1$

(a) $P(R) = P(R|L_1)P(L_1) + P(R|L_2)P(L_2) + P(R|L_3)P(L_3) + P(R|L_4)P(L_4)$

$= 0.4 \times 0.2 + 0.3 \times 0.1 + 0.2 \times 0.5 + 0.1 \times 0.2 = 0.08 + 0.03 + 0.1 + 0.02$

$= 0.23$

(b) $P(L_2|R) = \frac{0.3 \times 0.1}{0.23} = \frac{0.03}{0.23} = \frac{3}{23}$

$P(L_1|R) = \frac{8}{23}$ $P(L_4|R) = \frac{2}{23}$

$P(L_3|R) = \frac{10}{23} \rightarrow L_3$ is the most likely rate.

Q2 $P(X=0) = c\binom{2}{0}\binom{3}{3} = 3c = c$

$P(X=1) = c\binom{2}{1}\binom{3}{2} = 6c$ $10c = 1$, $c = \frac{1}{10}$

$P(X=2) = c\binom{2}{2}\binom{3}{1} = 3c$

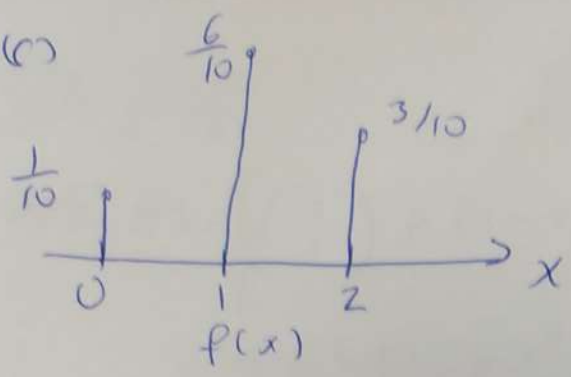
$P(X=0) = \frac{1}{10}$ $P(X=1) = \frac{6}{10}$ $P(X=2) = \frac{3}{10}$

(b) $E[X] = 0 \times \frac{1}{10} + 1 \times \frac{6}{10} + 2 \times \frac{3}{10} = \frac{12}{10} = 1.2$

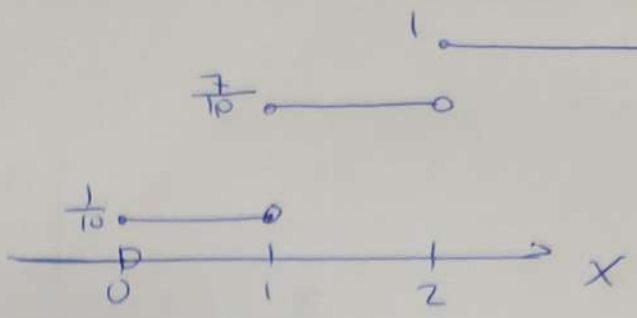
$E[X^2] = 0^2 \times \frac{1}{10} + 1^2 \times \frac{6}{10} + 2^2 \times \frac{3}{10} = \frac{18}{10}$

$Var(X) = 1.8 - (1.2)^2 = 0.36$

(c)



f(x)



F(x)

(d) $Y = 2X + 1$

$E[Y] = 2E[X] + 1 = 2 \times 1.7 + 1 = 3.4$

$Var(Y) = 4 Var(X) = 4 \times 0.36 = 1.44$

Q3 $p = 0.001$ $\lambda = np = 1.9$

$n = 1900$

$$P(X=0) = \frac{e^{-1.9}(1.9)^0}{0!} = 0.1496$$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= e^{-1.9} + e^{-1.9}(1.9) + \frac{e^{-1.9}(1.9)^2}{2}$$

$$= e^{-1.9}(1 + 1.9 + 1.8050) = e^{-1.9} \times 4.7050$$

$$= 0.7037$$

$$P(X=4) = \frac{e^{-1.9}(1.9)^4}{4!} = 0.0812$$

Q4 $p = \frac{1}{6}$

$$P(X=8) = \left[\binom{7}{1} p (1-p)^6 \right] p$$

$$= 7 p^2 (1-p)^6 = 7 \times \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^6 = 0.0651$$

$$P(Y=8) = (1-p)^7 p$$

$$= \left(\frac{5}{6}\right)^7 \times \frac{1}{6} = 0.0466$$